

Taxonomy of Multimodal Time Series Models

Multimodal Time Series Paradigm (Data, Fusion, Role, Task, Domain)

1. Modality Pairing

- TS + Text (Reports, News, Prompts)
- TS + Vision (Plots, Spectrograms, MAE)
- TS + Spikes / SNN (SpikySpace, TS-LIF)
 - TS + Audio (MTSA-SNN, Speech)
 - TS + Physics / PDEs (PhysDGM)
- TS + Spatio-Temporal Graphs (STReasoner)

2. Fusion Architecture

- Patch Reprogramming (Time-LLM, OFA)
- Spiking State Space / LIF (SpikySpace)
- Physics-Constrained Diffusion (PhysDGM)
- Continuous Neural CDE / SSM (SOTER, DeMa)
 - Spatial-Aware RL Policy (STReasoner)
 - Multi-Agent VLM Swarm (MAS4TS)

3. Non-TS Role

- Auxiliary Context / Condition
- Physical Conservation Law Residual
- Event-Driven Neuromorphic Trigger
 - Conformal Calibration Anchor
- Multi-Agent Supervisor / Tool Executor
 - Metric Alignment Target (TRACE)

4. Downstream Tasks

- Multimodal Forecasting (Point / Conformal)
 - Counterfactual Disaster Simulation
 - Micro-Watt Edge Anomaly Detection
- Spatio-Temporal Reasoning & Causal QA
 - Cross-Modal Retrieval (TRACE)
- Earth System Weather / Planetary Grids

Application Domains: Healthcare & EHR • Meteorology & Climate • Finance & Markets • Energy & Smart Grids • Geophysics & Bioacoustics